

|Implementation of TinyML for water classification by turbidity

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1. Introduction

Water quality is a critical indicator of environmental health, impacting human well-being, ecosystem sustainability, and economic development. One of the key parameters used to assess water quality is turbidity, which measures cloudiness caused by suspended particles such as sediment, clay, silt, and algae. High turbidity levels often indicate pollution, erosion, or other environmental disturbances that affect aquatic ecosystems and the usability of water for human consumption and other purposes [1].

Traditional methods of turbidity measurement rely on specialized instruments like nephelometers and turbidimeters, which work by measuring the scattering of light through water samples. While these methods provide accurate results, they come with significant limitations, including high equipment costs (ranging from \$500 to over \$5000), the need for laboratory settings, trained personnel, and the inability to offer continuous, real-time monitoring [2]. These challenges make it difficult to deploy such systems in remote locations and resource-constrained environments, where water quality monitoring is most needed.

Recent advancements in computer vision, deep learning, and IoT technologies have opened up new possibilities for measuring turbidity more cost-effectively. Image-based turbidity assessment leverages commodity hardware like smartphones and low-cost cameras, combined with AI algorithms, to provide accessible solutions [3], [4]. Furthermore, the integration of AIoT technologies

enables the development of distributed sensor networks that allow for continuous monitoring, real-time data transmission, and automated analysis [5].

Despite the promising developments, there are still gaps in existing systems. Many current solutions focus primarily on remote sensing, rather than in-situ monitoring, and are often not deployable in resource-limited areas. Additionally, there is insufficient integration of AIoT capabilities for real-time distributed monitoring, and a lack of comprehensive evaluations under varying environmental conditions. This work aims to address these challenges by presenting an AIoT-based system for sediment recognition in water, offering contributions such as an integrated AIoT architecture, a cost-effective implementation, comprehensive evaluations of CNN architectures, and practical deployment guidelines [6][7][8][9][10][11][12].

2. Proposal of Solution (System Architecture and General Details)

The proposed water turbidity classification system uses a three-tier architecture consisting of data acquisition, edge processing, and cloud analytics layers. In the data acquisition layer, a camera module captures images of water samples, which are taken under control or ambient lighting conditions. These images are processed using an optimized deep learning model deployed on an edge device, such as an Arduino-compatible microcontroller, to classify turbidity levels in real-time. The results are transmitted to the cloud via

Wi-Fi, where the ThingSpeak platform stores and visualizes the historical data, enabling trend analysis and alerting [7].

By performing inference at the edge, the system reduces the need for bandwidth by sending only classification labels, rather than raw images. This also lowers cloud processing costs and ensures functionality during network interruptions. The architecture is scalable, suitable for deployment in standalone monitoring stations or large-scale sensor networks, with the flexibility to adapt to different use cases.

The hardware stack utilizes Arduino-compatible microcontrollers such as the ESP32-CAM, which is capable of running TensorFlow Lite models. CMOS camera modules capture RGB images, and data augmentation techniques are applied to improve model generalization [14]. Communication modules like Wi-Fi, cellular, or LoRaWAN are used depending on the deployment context and network availability [15], [16].

The software stack leverages Edge Impulse Studio for dataset management, model training, and deployment. It integrates TensorFlow Lite for optimized inference on edge devices. The deployment process includes compiling and uploading firmware through the Arduino IDE, where the model processes the images, classifies the turbidity levels, and sends the results to ThingSpeak for visualization and monitoring [13].

3. Implementation

The proposed water turbidity classification system utilized the XIAO ESP32-S3 microcontroller with an integrated camera to capture 614 photographs of water samples against a black and white background. The camera's position was varied to ensure a range of capture angles, and turbidity levels were corroborated using NTU measurements. The classification was based on NTU values, where turbidity ranging from 0 to 5 NTU was considered "Excellent", from 5 to 50 NTU as "Moderate", and anything over 50 NTU as "Low". NTU readings were taken using SenseCraft AI, and this data was subsequently uploaded to the Edge Impulse platform for model development.

In preparation for model training, the dataset was split into an 80/20 ratio, with approximately 491 images allocated for training and 123 images for testing. Edge Impulse Studio served as the primary development environment, providing an integrated platform for dataset management, model configuration, and training. The images were preprocessed by resizing them to 96×96 pixels and converting them to grayscale, reducing the computational load while ensuring the model could focus on intensity variations relevant to turbidity. Normalization was applied to the grayscale images, scaling pixel values to a range of [0, 1], which improved the stability and speed of model training.

For model development, MobileNetV2 was selected due to its efficiency for mobile and embedded vision applications. Using transfer learning, the model was initialized with pre-trained weights from ImageNet. The model's architecture was adapted to perform a four-class classification, distinguishing between Low, Moderate and Excellent turbidity levels, as well as "Empty" ensure the ML model would make a background distinction. Edge Impulse streamlined the transfer learning process by automating the weight initialization, layer freezing, and classification head replacement.

Model training was accelerated using GPU resources on Edge Impulse's cloud-based infrastructure, with a configuration of 30 epochs to balance training time and convergence quality. A learning rate of 0.00005 was set, and data augmentation techniques such as rotation, flipping, and brightness adjustments were applied to the training data. These augmentations helped the model generalize better to variations in camera angle, lighting conditions, and positioning, further improving its robustness.

Once the model was trained, post-training optimization was carried out using Edge Impulse's EON Compiler. This optimization process included quantization, operator fusion, and memory layout optimizations to reduce the model's size and accelerate inference, ensuring it could be deployed on resource-constrained hardware. The final model was converted into optimized Arduino library, which was then deployed to the ESP32-S3 using

Arduino IDE. The TensorFlow Lite Micro framework provided a lightweight inference engine, suitable for real-time classification on embedded devices.

The system's deployment involved capturing images using the camera, processing them with the model, and sending the classification results to the ThingSpeak platform via HTTP POST requests. ThingSpeak enabled real-time data visualization, allowing for monitoring of water turbidity trends through customizable dashboards and charts. The platform's analytics capabilities, including anomaly detection and trend forecasting, enhanced the system's ability to provide actionable insights into water quality.

4. Results

4.1. Training Performance

The MobileNetV2 model achieved an accuracy of 80.6% during the training phase after 30 epochs of GPU-accelerated training on the 491-image training set. This performance indicates that the model effectively learned discriminative features for turbidity classification, with a loss of 0.62 using categorical cross-entropy. This training performance is notable for a four-class classification task and suggests that the model is capturing meaningful patterns in the images, correctly classifying approximately four out of five training samples.

	BAJA	EXCELENTE	MODERADA	VACIO
BAJA	79.5%	0%	20.5%	0%
EXCELENTE	0%	75%	16.7%	8.3%
MODERADA	15.2%	3.0%	81.8%	0%
VACIO	14.3%	0%	0%	85.7%
F1 SCORE	0.81	0.82	0.77	0.89

Figure 1. Confusion matrix from training phase.

The training confusion matrix showed that the model correctly identified 79.5% of “Low” images, 75% of “Excellent” images, 81.8% of “Moderate” turbidity images, and 85.7% of “Empty” (no water) images. There was a significant confusion between low and moderate turbidity classes, with 20.5% of “Low” turbidity samples misclassified as “Moderate” and 16.7% of moderate samples misclassified as “Low”. This suggests that further data collection or refinement of class definitions

could help reduce this overlap. Additionally, the “Excellent” category had 16.7% misclassified as “Moderate”, highlighting the potential challenge of classifying water clarity near the upper end of the turbidity scale.

While there were some misclassifications, especially for the “Moderate” and “Low” turbidity categories, the model was able to generalize reasonably well across the different turbidity levels. The loss value of 0.62 suggests moderate confidence in the predictions, indicating room for improvement with further training or additional data.

4.2. Test Performance

The test performance confusion matrix provides valuable insights into the model's performance for each turbidity category. When tested on a separate set of 123 images, the model achieved an accuracy of 70.97%, demonstrating moderate generalization capabilities. This test accuracy is approximately 10 percentage points lower than the training accuracy, which is typical for deep learning models, particularly when training data is limited. The confusion matrix for the test set indicated that the model correctly classified 62.2% of “Low” samples, 57.1% of “Excellent” turbidity samples, 76.3% of “Moderate” turbidity samples, and 95% of “Empty” samples. The test set accuracy suggests the model is effective in real-world water quality monitoring but may require additional fine-tuning to improve classification reliability for edge cases.

	BAJA	EXCELENTE	MODERADA	VACIO	UNCERTAIN
BAJA	62.2%	0%	15.6%	2.2%	20%
EXCELENTE	0%	57.1%	0%	14.3%	28.6%
MODERADA	0%	5.3%	76.3%	2.6%	15.8%
VACIO	0%	0%	0%	95%	5%
F1 SCORE	0.77	0.69	0.78	0.86	

Figure 2. Confusion matrix from testing phase

The weighted average F1 score across classes was 0.77, showing a balanced performance across the categories. In this testing phase, the model demonstrated lower precision for the “Moderate”, “Low”, and “Excellent” classifying them as “Uncertain”. The high accuracy for “Empty” samples is likely due to the distinct visual features

of containers without water, which are easily distinguishable from water-filled samples.

4.3. Deployment into ESP32-S3 and ThingSpeak

The final stage is the deployment of the ML model to the ESP32-S3. On startup, the chip initializes the camera, display, and WiFi connection before reserving the necessary memory to run inference. Then, in a continuous loop, the camera captures a frame, converts and resizes it to match the model's input format, and runs the neural network directly on the chip. The resulting probabilities for each class (Low, Excellent, Moderate, and Empty) are displayed on both the Serial Monitor of the Arduino IDE and the integrated Display on the Expansion Board. Finally, every 16 seconds the inference results are transmitted via WiFi onto ThingSpeak. Below some images are presented to demonstrate the functionality of the deployment of the TinyML project.



Image 1. Demonstration of the deployment of the TinyML project.

5. Limitations and future prospects

Several limitations constrain the current implementation of the water turbidity classification system, which suggest areas for future improvement:

- **Dataset Size and Diversity:** The dataset, consisting of only 614 images, is relatively small for deep learning applications, which may limit the model's ability to generalize well to different real-world conditions. Increasing the size of the dataset and incorporating a more diverse range of samples—considering factors like lighting conditions, backgrounds, water sources, and turbidity levels—would enhance the model's robustness and improve performance when deployed in various settings.
- **Class Definition Ambiguity:** The current categorization system for turbidity, which divides the data into four classes (Low, Moderate, Empty, Excellent), relies on subjective thresholds, particularly between adjacent turbidity levels. More precise definitions, possibly supported by quantitative measurements such as NTU (Nephelometric Turbidity Units), would increase the consistency and accuracy of class labels, leading to a more reliable model.
- **Lighting Condition Sensitivity:** Despite using data augmentation techniques, the model may still be sensitive to variations in lighting that were not adequately represented in the training data. Differences in lighting, such as bright or dim environments, could cause turbidity levels to appear different, leading to misclassifications. Future improvements could involve more diverse training under various lighting conditions or the implementation of preprocessing techniques that adjust for these variations.
- **Grayscale Limitation:** The decision to convert images to grayscale simplifies the model's computational load but also removes potential valuable color information, which could help the model distinguish between certain turbidity levels, especially when dealing with colored contaminants or algae. Future work could explore the potential benefits of including color information, weighing

the trade-off between accuracy and computational efficiency.

- **Hardware Constraints:** The use of Arduino-compatible microcontrollers imposes certain limitations on memory and processing capabilities, forcing the model to undergo aggressive compression, which could impact accuracy.
- **Limited Field Validation:** The results reported thus far have been based on a controlled testing environment. For the system to be fully validated, it must be deployed in the field under varied environmental conditions and with real-world water samples. Such extensive field validation is crucial to assess the system's reliability in practice and to identify any potential failure modes.

6. Conclusion

The project demonstrates that it is technically feasible to implement a water turbidity classification system using optimized deep learning architectures (MobileNetV2) on low-cost microcontrollers like the XIAO ESP32-S3. The project demonstrates that it is technically feasible to implement a water turbidity classification system using optimized deep learning architecture (MobileNetV2) on low-cost microcontrollers like the XIAO ESP32-S3. This TinyML approach offers a highly accessible alternative to traditional measurement methods—such as nephelometers and turbidimeters—which can cost between \$500 and over \$5,000. By significantly reducing equipment costs, this system enables real-time water quality monitoring in remote or resource-constrained environments where such data is most needed.

The implemented three-tier architecture (data acquisition, edge processing, and cloud analytics) ensures high operational efficiency. By performing inference directly on the edge device, the system optimizes bandwidth by transmitting only classification labels to the ThingSpeak platform rather than raw images. This Edge Computing capability not only lowers cloud processing costs

but also maintains basic functionality during network interruptions, making the system scalable for both standalone stations and large sensor networks.

Regarding model performance, the system achieved a training accuracy of 80.6% and a test accuracy of 70.97%, indicating moderate generalization capabilities. Regarding model performance, the system achieved a training accuracy of 80.6% and a test accuracy of 70.97%, indicating good generalization capabilities. The model was exceptionally effective at identifying "empty" containers with 95% accuracy due to their distinct visual features. The model was exceptionally effective at identifying "Empty" containers with 95% accuracy due to their distinct visual features. However, a significant overlap was observed between "low" and "moderate" turbidity levels, with approximately 20.5% of low-turbidity samples being misclassified as moderate. However, a significant overlap was observed between "Low" and "Moderate" turbidity levels, with approximately 20.5% of low-turbidity samples being misclassified as moderate. This suggests that refining class definitions or increasing the volume of training data is necessary to improve reliability for borderline cases.

Finally, several areas for future improvement were identified to enhance the system's robustness. The current dataset of 614 images is relatively small and expanding it to include diverse lighting conditions and backgrounds is crucial for real-world deployment. While using grayscale images reduced the computational load, future work could explore incorporating color information to better distinguish between algae or specific contaminants. Lastly, moving from controlled laboratory settings to extensive field validation is essential to confirm the system's reliability under varied environmental conditions.

7. References

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